SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA1613 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

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QUESTIONS: 05 **Total Marks:50**

Annexure A

MOCK INTERVIEW

Criteria	Technical Knowledge *(2.5)*	Problem Solving *(2.5)*	Analytical Thinking *(1.5)*	***Communication(1.5)***	Confidence *(1)*	Response o t 1 Questions ( )	***Total(10)***
Weightage	*25%*	25%	15%	15%	10%	*10%*	*100%*
Q1							
Q2							

Code of Conduct:

I **G.MOHAMMAD SHAREEF/(192311288)***(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:G.MOHAMMAD SHAREEF

RUBRICS FOR CALCULATION:

Criteria	Weightag e	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)

Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Annexure A

Mock Interview – Sample Questions (5 Questions)

Q1. Dataset: Retail Transactions

TID	Items
T1	Milk, Bread
T2	Milk, Butter

T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Milk, Eggs

Question:
Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example.

Q2. Dataset: Transaction Data

TID Items

T1 A,	B
T2 B,	C
T3 A,	C
T4 A, B,	C
T5 B,	C

Question:
Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used.

Q3. Dataset: Transaction Data

TID Items

T1	A, B
T2	A
T3	B
T4	A, B
T5	A

Question:
Explain support and confidence using the dataset and describe their importance in association rule mining.

Q4 . Dataset: Multilevel Data

TID	**Items**
T1	Electronics, Mobile
T2	Electronics, Laptop
T3	Electronics, Mobile

T4	Electronics, Mobile
T5	Electronics, Laptop

Question:
Explain multilevel association rules using the dataset and compare them with multidimensional

Q5. Dataset: Large Transaction Sample

TID	**Items**
T1	A, B, C
T2	A, B
T3	B, C
T4	A, C
T5	A, B, C

Question:
Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues.

Q1. Association Rule Mining – Retail Transactions

Given Dataset

TID	Items
T1	Milk, Bread
T2	Milk, Butter
T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Milk, Eggs

The question asks how association rule mining can improve retail sales and to provide a practical example.

Step 1: What is Association Rule Mining?

Association rule mining is a data-mining technique used to discover relationships between items frequently purchased together.

A rule is represented as:

X → Y

Example:

Milk → Bread

This means customers who purchase Milk are likely to purchase Bread.

Step 2: Calculate Support

Formula:

$$Support(X) = \frac{Number\ of\ transactions\ containing\ X}{Total\ number\ of\ transactions}$$

There are **5 transactions**.

For Milk:

Milk appears in:

- T1
- T2
- T4
- T5

Therefore:

$$Support(Milk) = \frac{4}{5} = 0.8 = 80\%$$

For Bread:

Bread appears in:

- T1
- T3
- T4

Therefore:

$$Support(Bread) = \frac{3}{5} = 60\%$$

Step 3: Calculate Support for Milk and Bread

Milk and Bread occur together in:

- T1
- T4

Therefore:

$$Support(ilkBread) = \frac{2}{5} = 0.4 = 40\%$$

Step 4: Calculate Confidence

For the rule:

Milk → Bread

Formula:

$$Confidence(Milk{\rightarrow}Bread) = \frac{Support(ilkBread)}{Support(Milk)}$$

Substitution:

$$= \frac{0.4}{0.8} = 0.5 = 50\%$$

Interpretation

The confidence of **50%** means that **50% of transactions containing Milk also contain Bread**.

Practical Retail Example:

Suppose a supermarket discovers:

Milk → Bread

with sufficiently high support and confidence.

The store can:

- Place bread near the milk section.
- Give a combo offer for Milk + Bread.
- Recommend Bread when customers buy Milk online.
- Keep sufficient stock of both products.
- Create targeted promotions.

Final Answer:

Association rule mining helps retailers identify **frequently purchased product combinations**. In this dataset, Milk and Bread occur together in 2 out of 5 transactions, giving **40% support**, while the rule Milk → Bread has **50% confidence**. Therefore, the retailer can use this relationship for product placement, recommendations, and combo offers.

Q2. Apriori vs FP-Growth

Given Dataset

TID	**Items**
T1	A, B
T2	B, C
T3	A, C
T4	A, B, C
T5	B, C

The PDF asks to explain the differences between Apriori and FP-Growth and when each should be used.

Apriori Algorithm:

Apriori finds frequent itemsets using the **Apriori property**:

If an itemset is frequent, all of its subsets must also be frequent.

Step 1: Find 1-itemsets

A occurs in:

- T1
- T3
- T4

$$Support(A) = \frac{3}{5} = 60\%$$

B occurs in:

- T1
- T2
- T4
- T5

$$Support(B) = \frac{4}{5} = 80\%$$

C occurs in:

- T2
- T3
- T4
- T5

$$Support(C) = \frac{4}{5} = 80\%$$

Therefore:

Item	Count	Support
A	3	60%
B	4	80%
C	4	80%

Step 2: Find 2-itemsets

AB

AB appears in:

- T1
- T4

$$Support(AB) = \frac{2}{5} = 40\%$$

AC

AC appears in:

- T3
- T4

$$Support(AC) = \frac{2}{5} = 40\%$$

BC

BC appears in:

- T2
- T4
- T5

$$Support(BC) = \frac{3}{5} = 60\%$$

Step 3: Find 3-itemset

ABC occurs only in:

- T4

Therefore:

$$Support(ABC) = \frac{1}{5} = 20\%$$

If minimum support is **20%**, ABC is frequent.

FP-Growth:

FP-Growth stands for **Frequent Pattern Growth**.

Instead of repeatedly generating candidate itemsets like Apriori, FP-Growth:

1. Scans the database.

2. Finds frequent items.
3. Builds an **FP-tree**.
4. Compresses transactions.
5. Mines frequent patterns directly from the tree.

Major Difference:

Feature	**Apriori**	**FP-Growth**
Candidate generation	Yes	No
Database scans	Multiple	Usually two major scans
Data structure	Candidate sets	FP-tree
Speed	Slower for large data	Generally faster
Memory	Can require many candidates	Compact tree
Best suited for	Small/moderate datasets	Large transaction datasets

When to use Apriori?

Use Apriori when:

- Dataset is relatively small.
- Algorithm simplicity is important.
- Number of candidate itemsets is manageable.
- Easy interpretation is required.

When to use FP-Growth?

Use FP-Growth when:

- Dataset is large.
- There are many transactions.
- Many frequent itemsets exist.
- Apriori candidate generation becomes expensive.

Final Answer:

Apriori uses candidate generation and repeated database scans, whereas **FP-Growth** constructs an FP-tree and mines patterns without explicitly generating candidates. Therefore, Apriori is suitable for smaller datasets, while FP-Growth is generally preferred for large transaction databases.

Q3. Support and Confidence

Given Dataset

TID	Items
T1	A, B
T2	A
T3	B
T4	A, B
T5	A

The question specifically asks to explain support and confidence using this dataset.

There are:

$$N = 5$$

transactions.

Part A: Support:

Support of A:

A occurs in:

- T1
- T2
- T4
- T5

Number of occurrences:

$$4$$

Formula:

$$Support(A) = \frac{4}{5} = 0.8 = 80\%$$

Support of B:

B occurs in:

- T1
- T3
- T4

Therefore:

$$Support(B) = \frac{3}{5} = 0.6 = 60\%$$

Support of A and B:

A and B occur together in:

- T1
- T4

Therefore:

$$Support(B) = \frac{2}{5} = 0.4 = 40\%$$

Part B: Confidence:

Consider the rule:

$$A \rightarrow B$$

Formula:

$$Confidence(A \rightarrow B) = \frac{Support(B)}{Support(A)}$$

Substitution:

$$= \frac{0.4}{0.8} = 0.5 = 50\%$$

Interpretation

This means:

50% of transactions containing A also contain B.

Confidence of B → A

Formula:

$$Confidence(B \rightarrow A) = \frac{Support(B)}{Support(B)}$$

Substitution:

$$= \frac{0.4}{0.6} = 0.6667 = 66.67\%$$

Therefore:

- A → B = **50% confidence**
- B → A = **66.67% confidence**

Notice that the two confidence values are different even though they involve the same two items.

Why Support is Important?

Support tells us **how frequently an itemset occurs in the entire database**.

Higher support means the pattern occurs more often.

For A and B:

$$Support(B) = 40\%$$

So 40% of all transactions contain both A and B.

Why Confidence is Important?

Confidence measures how strongly the occurrence of one item is associated with another.

For:

$$A \rightarrow B$$

confidence is:

$$50\%$$

Therefore, among transactions containing A, half also contain B.

Final Answer:

Support measures the **frequency of an itemset**, while confidence measures the **reliability of an association rule**. Both are important because support identifies useful frequent patterns and confidence determines how reliable the resulting rules are.

Q4. Multilevel Association Rules

Given Dataset

TID	Items
T1	Electronics, Mobile
T2	Electronics, Laptop
T3	Electronics, Mobile
T4	Electronics, Mobile
T5	Electronics, Laptop

The question asks about multilevel association rules and comparison with multidimensional association rules.

What is a Multilevel Association Rule?

Multilevel association mining discovers relationships at **different levels of abstraction**.

For example:

Level 1:

$$Electronics \rightarrow Mobile$$

Level 2:

$$Smartphone \rightarrow Android$$

The first rule is more general, while the second is more specific.

Mathematical Example:

Electronics appears in all 5 transactions.

$$Support(Electronics) = \frac{5}{5} = 100\%$$

Mobile occurs in:

- T1

- T3
- T4

Therefore:

$$Support(Mobile) = \frac{3}{5} = 60\%$$

Laptop occurs in:

- T2
- T5

Therefore:

$$Support(Laptop) = \frac{2}{5} = 40\%$$

Rule: Electronics → Mobile

Electronics and Mobile occur together in:

- T1
- T3
- T4

Therefore:

$$Support(lectronicsMobile) = \frac{3}{5} = 60\%$$

Confidence:

$$Confidence(Electronics \rightarrow Mobile) = \frac{Support(lectronicsMobile)}{Support(Electronics)}$$

Substitution:

$$= \frac{0.60}{1.00} = 60\%$$

So:

$$Confidence = 60\%$$

Multilevel vs Multidimensional Association:

Multilevel Association	Multidimensional Association
Uses different levels of abstraction	Uses multiple attributes/dimensions
Example: Electronics → Mobile	Example: Age → Product
Uses product hierarchy	Uses attributes such as age, location, income
General-to-specific analysis	Cross-dimensional analysis

Example

Multilevel:

$$Electronics \rightarrow Mobile$$

Multidimensional:

$$Age = 18 - 25 \land Location = Chennai \rightarrow Mobile$$

Final Answer:

Multilevel association rules discover relationships at different abstraction levels, such as **Electronics → Mobile**. Multidimensional association rules involve multiple attributes such as customer age, location, income, and product. Multilevel rules focus on **hierarchical levels**, whereas multidimensional rules focus on **multiple dimensions**.

Q5. Challenges in Mining Large Datasets

Given Dataset

TID	Items
T1	A, B, C
T2	A, B
T3	B, C
T4	A, C
T5	A, B, C

The PDF asks to discuss challenges in mining large datasets and explain how scalable methods address them.

Step 1: Calculate Item Supports

A

A appears in:

- T1
- T2
- T4
- T5

$$Support(A) = \frac{4}{5} = 80\%$$

B

B appears in:

- T1
- T2
- T3
- T5

$$Support(B) = \frac{4}{5} = 80\%$$

C

C appears in:

- T1
- T3
- T4
- T5

$$Support(C) = \frac{4}{5} = 80\%$$

Step 2: Two-Itemsets

AB

AB appears in:

- T1
- T2
- T5

$$Support(AB) = \frac{3}{5} = 60\%$$

AC

AC appears in:

- T1
- T4
- T5

$$Support(AC) = \frac{3}{5} = 60\%$$

BC

BC appears in:

- T1
- T3
- T5

$$Support(BC) = \frac{3}{5} = 60\%$$

Step 3: Three-Itemset

ABC appears in:

- T1
- T5

Therefore:

$$Support(ABC) = \frac{2}{5} = 40\%$$

So the frequent patterns include:

Itemset	Support
A	80%
B	80%
C	80%
AB	60%
AC	60%
BC	60%
ABC	40%

Challenges in Large-Scale Association Mining

1. Huge Dataset Size:

Large databases may contain millions or billions of transactions.

This increases:

- Processing time
- Memory requirements
- Storage requirements

2. Candidate Explosion:

In Apriori, the number of possible itemsets can grow rapidly.

For **n items**, the number of possible non-empty itemsets is:

$$2^n - 1$$

For example, if:

$$n = 10$$

then:

$$2^{10} - 1 = 1023$$

possible non-empty itemsets exist.

If:

$$n = 20$$

then:

$$2^{20} - 1 = 1,048,575$$

This demonstrates why candidate generation can become expensive.

3. Multiple Database Scans:

Apriori repeatedly scans the transaction database for candidate support calculation.

For very large databases, this causes significant I/O and processing overhead.

How Scalable Methods Solve These Problems:

FP-Growth:

FP-Growth avoids explicit candidate generation and uses an **FP-tree** to compress the database.

This reduces:

- Candidate generation
- Repeated database scans
- Memory and processing overhead

Parallel and Distributed Mining:

Large datasets can be divided among multiple machines.

For example:

$$Dataset = D_1 + D_2 + D_3 + D_4$$

Different processors can work on different portions simultaneously.

This reduces overall execution time.

Data Partitioning:

The dataset can be divided into smaller partitions.

Instead of processing:

$$10,000,000$$

transactions at once, the system can process manageable partitions.

Sampling:

A representative sample can sometimes be mined instead of the complete database, reducing computational requirements.

Final Interview Answer:

Mining large datasets is challenging because of **huge data volume, candidate itemset explosion, repeated database scans, memory requirements, and processing time**. For example, with $n$items, the number of possible non-empty itemsets can reach $2^n - 1$. Scalable methods such as **FP-Growth, parallel processing, distributed mining, partitioning, and sampling** reduce computation and improve mining efficiency.